

Curtin University, Semester 2, 2026
ECOM5005
Tutorial 7A Detailed Solutions

The Normal Distribution and Other Continuous Distributions - classroom-ready worked solutions

Learning Outcomes

1. Calculate and interpret normal probabilities and percentiles.
2. Use multiple diagnostics to assess approximate normality.
3. Solve uniform and exponential waiting-time problems.
4. Apply continuity correction in the normal approximation to a binomial distribution.
5. Carry out a complete beginner-friendly Excel analysis of a real-world continuous dataset.

PART 1 - Detailed worked solutions

Question 1. Grocery-delivery service levels

Question

X is normally distributed with mean $\mu = 42$ minutes and standard deviation $\sigma = 8$ minutes. Find the requested probabilities, the 90th percentile, and the expected number of deliveries exceeding 60 minutes.

Detailed Solution

Step 1: Standardise any delivery time using $Z = (X - \mu)/\sigma$.

(a) For 35 minutes: $Z = (35 - 42)/8 = -0.875$. Therefore $P(X < 35) = P(Z < -0.875) = 0.1908$. In Excel: `=NORM.DIST(35,42,8,TRUE)`.

(b) $P(35 < X < 55) = P(X < 55) - P(X < 35)$. The upper Z-score is $(55 - 42)/8 = 1.625$. Excel: `=NORM.DIST(55,42,8,TRUE)-NORM.DIST(35,42,8,TRUE) = 0.7571`.

(c) The 90th percentile is the value x satisfying $P(X \leq x) = 0.90$. Excel: `=NORM.INV(0.90,42,8) = 52.2524` minutes.

(d) $P(X > 60) = 1 - \text{NORM.DIST}(60,42,8,TRUE) = 0.01222$. For 200 deliveries, expected count = $200 \times 0.01222 = 2.44$ deliveries.

Interpretation

About 19.1% of deliveries finish within 35 minutes; about 75.7% fall between 35 and 55 minutes. A 90%-service threshold is about 52.3 minutes. On an evening with 200 deliveries, roughly 2 to 3 deliveries would be expected to exceed 60 minutes.

Answer Summary

(a) 0.1908 (b) 0.7571 (c) 52.25 minutes (d) 2.44 expected deliveries, i.e. roughly 2-3.

Question 2. Warehouse robot cycle times

Detailed Solution

The lecture gives several diagnostics for normality. For a normal distribution, mean and median should be similar; IQR should be about 1.33σ ; range is often roughly 6σ ; about 68.26%, 80%, and 95.44% should fall within $\pm 1\sigma$, $\pm 1.28\sigma$, and $\pm 2\sigma$; skewness and excess kurtosis should be near 0; and the normal probability plot should be approximately linear.

Here, mean 63.8 is noticeably above median 60.1, suggesting right-skewness. $IQR/\sigma = 16.2/15 = 1.08$, below 1.33. $Range/\sigma = 108/15 = 7.20$, above 6. The empirical percentages 74%, 88%, 96% are not disastrous individually, but skewness = 1.35 and kurtosis = 2.90 are far from zero. Most importantly, the Q-Q plot bends upward in the upper tail. Taken together, the diagnostics indicate a right-skewed, heavy-tailed distribution rather than a good normal approximation.

If a manager nevertheless assumes $N(63.8, 15^2)$: $Z = (90 - 63.8)/15 = 1.7467$, so $P(X > 90) = 1 - \Phi(1.7467) = 0.0404$. Excel: `=1-NORM.DIST(90,63.8,15,TRUE)`.

Interpretation

The model-based estimate is about 4.0%, but the observed right tail is heavier than a normal model suggests. Therefore the true probability of very long cycle times may be materially higher. This is exactly why normality should be checked before relying on normal-tail probabilities.

Question 3. Rideshare pickup promise

Detailed Solution

For $X \sim \text{Uniform}(a=4, b=12)$, the density is constant and interval probabilities are simply interval length divided by total length.

(a) $P(X < 6) = (6 - 4)/(12 - 4) = 2/8 = 0.25$.

(b) $P(7.5 < X < 10) = (10 - 7.5)/8 = 2.5/8 = 0.3125$.

(c) Mean $\mu = (a+b)/2 = (4+12)/2 = 8$ minutes. Standard deviation $\sigma = (b-a)/\sqrt{12} = 8/\sqrt{12} = 2.3094$ minutes.

(d) $P(X > 10) = (12 - 10)/8 = 0.25$. Expected credits among 500 rides = $500 \times 0.25 = 125$. Expected cost = $125 \times \$5 = \625 .

Answer Summary

(a) 0.25 (b) 0.3125 (c) $\mu = 8.00$ minutes, $\sigma = 2.3094$ minutes (d) 125 credits; expected cost = \$625.

Question 4. Cybersecurity incident arrivals

Detailed Solution

For an exponential waiting-time model with rate $\lambda = 3$ per hour, $P(X < A) = 1 - e^{-\lambda A}$. Time must use the same unit as λ .

(a) 10 minutes = $10/60 = 1/6$ hour. $P(X < 10 \text{ min}) = 1 - e^{-3(1/6)} = 1 - e^{-0.5} = 0.3935$.

(b) $P(X > 30 \text{ min}) = e^{-\lambda A} = e^{-3(0.5)} = e^{-1.5} = 0.2231$.

(c) Set $0.90 = 1 - e^{-3t}$. Then $e^{-3t} = 0.10$, so $t = -\ln(0.10)/3 = 0.7675$ hours = 46.05 minutes.

Interpretation

There is about a 39.3% chance of another high-priority alert within 10 minutes and about a 22.3% chance of waiting more than 30 minutes. A 90% planning window is approximately 46.1 minutes: under the model, 90% of inter-alert times are shorter than this.

Question 5. Online-order returns

Detailed Solution

Let $X \sim \text{Binomial}(n=600, p=0.08)$.

(a) Check the approximation: $np=600 \times 0.08=48 \geq 5$ and $n(1-p)=600 \times 0.92=552 \geq 5$, so the lecture rule is satisfied. The approximating normal distribution has $\mu=np=48$ and $\sigma=\sqrt{np(1-p)}=\sqrt{(44.16)}=6.6453$.

(b) $X \geq 60$ becomes $X > 59.5$ after continuity correction. $Z=(59.5-48)/6.6453=1.7305$. Therefore $P(X \geq 60) \approx P(Z > 1.7305)=0.0418$.

(c) $40 \leq X \leq 55$ becomes $39.5 < X < 55.5$. The Z-scores are $(39.5-48)/6.6453=-1.2791$ and $(55.5-48)/6.6453=1.1286$. Thus $P(40 \leq X \leq 55) \approx \Phi(1.1286) - \Phi(-1.2791)=0.7700$.

(d) The binomial is discrete, while the normal model is continuous. The ± 0.5 continuity adjustment makes the continuous interval cover the full width of the relevant integer bars, improving the approximation.

Answer Summary

Normal approximation is appropriate. $\mu=48$, $\sigma=6.6453$. Approximate $P(X \geq 60)=0.0418$ and $P(40 \leq X \leq 55)=0.7700$.

PART 2 - Data exercise: Palmer Penguins

Dataset note

The supplied workbook contains 100 Adelie penguin records from the public Palmer Penguins dataset. The broader dataset contains measurements from 344 penguins and was collected by Dr. Kristen Gorman and Palmer Station, Antarctica LTER. The teaching workbook retains one record with missing numeric measurements so students see how Excel treats blanks. Source: <https://github.com/allisonhorst/palmerpenguins/blob/main/inst/extdata/penguins.csv>

Beginner Excel workflow - do these steps in order

Step 1: Open “Palmer_Penguins_Data.xlsx”. Click the Penguin_Data sheet. Do not sort or delete rows on this sheet. The headers are in row 5 and observations are in rows 6-105.

Step 2: Create your calculations on the Analysis_Workspace sheet. Bill Length is Penguin_Data!C6:C105; Bill Depth is D6:D105; Flipper Length is E6:E105; Body Mass is F6:F105. There is one blank numeric record. COUNT, AVERAGE, MEDIAN and STDEV.S automatically ignore blank cells.

Step 3: For descriptive statistics, type labels such as Count, Mean, Median, Mode, Standard deviation, Variance, Minimum, Q1, Q3, Maximum, Range, Skewness and Kurtosis down one column. In the adjacent column, enter formulas. Example for Bill Length: =COUNT(Penguin_Data!C6:C105), =AVERAGE(Penguin_Data!C6:C105), =MEDIAN(...), =MODE.SNGL(...), =STDEV.S(...), =VAR.S(...), =MIN(...), =QUARTILE.INC(...,1), =QUARTILE.INC(...,3), =MAX(...), =MAX(...)-MIN(...), =SKEW(...), =KURT(...). Repeat for columns D, E and F.

Step 4: To create the Bill Length histogram: select Penguin_Data!C6:C105, choose Insert > Insert Statistic Chart > Histogram. If Excel asks about blanks, simply leave the blank record unselected or allow Excel to ignore it. Add the title “Adelie Penguin Bill Length (mm)” and label the horizontal axis “Bill length (mm)”.

Step 5: For empirical-rule checks, calculate the bill-length mean and standard deviation first. For $\pm 1\sigma$, compute lower = mean-SD and upper = mean+SD. Then use =COUNTIFS(Penguin_Data!C6:C105,">="&lower_cell,Penguin_Data!C6:C105,"<="&upper_cell)/COUNT(Penguin_Data!C6:C105). Repeat using $1.28 \times \text{SD}$ and $2 \times \text{SD}$.

Step 6: To create the Q-Q plot: copy the 99 non-blank bill lengths to a new column and sort Smallest to Largest. Number them $i=1, \dots, 99$. In the next column calculate $p_i=(i-0.5)/99$. In the next column calculate theoretical Z with $=\text{NORM.S.INV}(p_i)$. Select the Z column and ordered bill-length column, then Insert > Scatter > Scatter with only markers. Add a Linear Trendline and tick "Display R-squared value on chart". A nearly straight pattern supports approximate normality.

Step 7: For fitted-normal probabilities, use the sample mean and sample standard deviation you calculated. $P(X < 36) = \text{NORM.DIST}(36, \text{mean_cell}, \text{sd_cell}, \text{TRUE})$. $P(37 \leq X \leq 41) = \text{NORM.DIST}(41, \text{mean_cell}, \text{sd_cell}, \text{TRUE}) - \text{NORM.DIST}(37, \text{mean_cell}, \text{sd_cell}, \text{TRUE})$. $P(X > 44) = 1 - \text{NORM.DIST}(44, \text{mean_cell}, \text{sd_cell}, \text{TRUE})$.

Step 8: For empirical probabilities, divide counts by the number of non-missing observations. Use $=\text{COUNTIF}(\text{Penguin_Data!C6:C105}, "<36")/\text{COUNT}(\text{Penguin_Data!C6:C105})$; $=\text{COUNTIFS}(\text{Penguin_Data!C6:C105}, ">=37", \text{Penguin_Data!C6:C105}, "<=41")/\text{COUNT}(\dots)$; and $=\text{COUNTIF}(\text{Penguin_Data!C6:C105}, ">44")/\text{COUNT}(\dots)$.

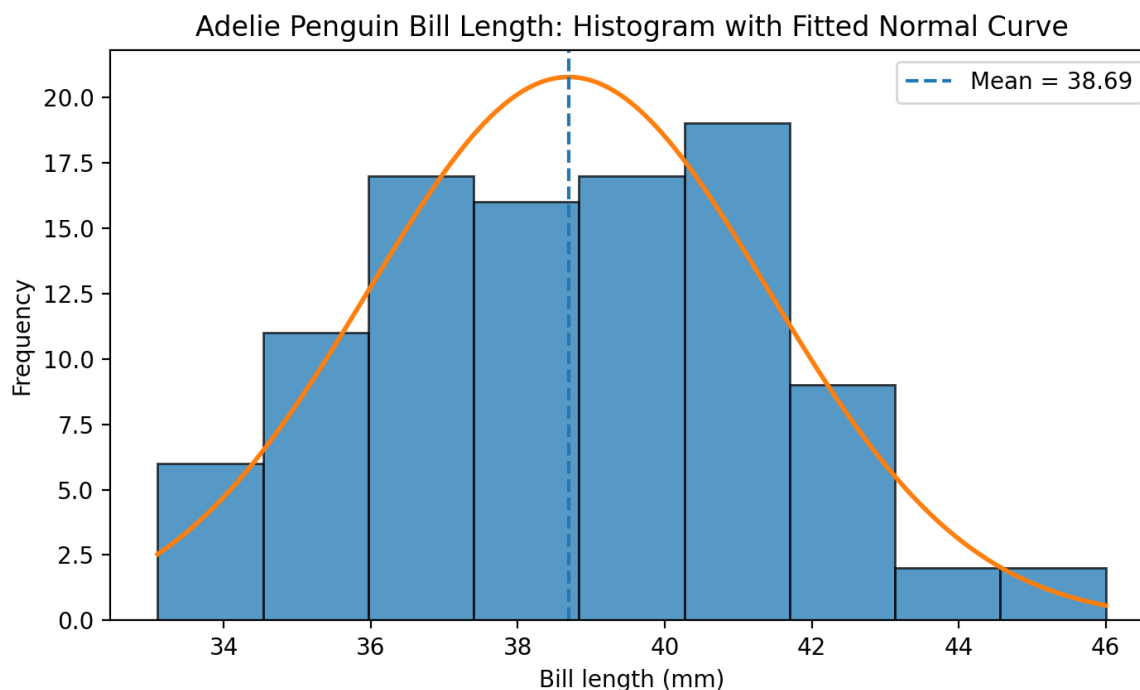
Step 9: For Z-scores, next to each non-blank bill length enter $=(\text{cell_with_bill_length} - \text{mean_cell})/\text{sd_cell}$ and fill down. Flag values with $|Z| > 2$ using $=\text{IF}(\text{ABS}(z_cell) > 2, \text{"Check"}, "")$. Do not delete a row merely because it is unusual; first consider whether it is a plausible genuine observation, a recording error, or a value from a different population.

Final descriptive-statistics results

Statistic	Bill Length	Bill Depth	Flipper Length	Body Mass
Count	99	99	99	99
Mean	38.6909	18.4768	188.8384	3719.4444
Median	38.8	18.5	190	3700
Mode	41.1	18.9	190	3800
Std. deviation	2.7252	1.1902	6.4566	450.8715
Variance	7.4268	1.4165	41.6879	203285.1474
Minimum	33.1	16.1	172	2850
Q1	36.5	17.75	185	3400
Q3	40.8	19.1	193.5	4025
Maximum	46.0	21.5	208	4700
Range	12.9	5.4	36	1850
Skewness	0.1962	0.4190	0.0746	0.1952
Kurtosis	-0.3321	0.1523	0.2698	-0.5936

Interpretation: all four numeric variables in this Adelie subset are fairly symmetric by skewness, although none is guaranteed to be exactly normal. Bill Length is the focus for the more detailed normality assessment.

Histogram and numerical normality checks for Bill Length



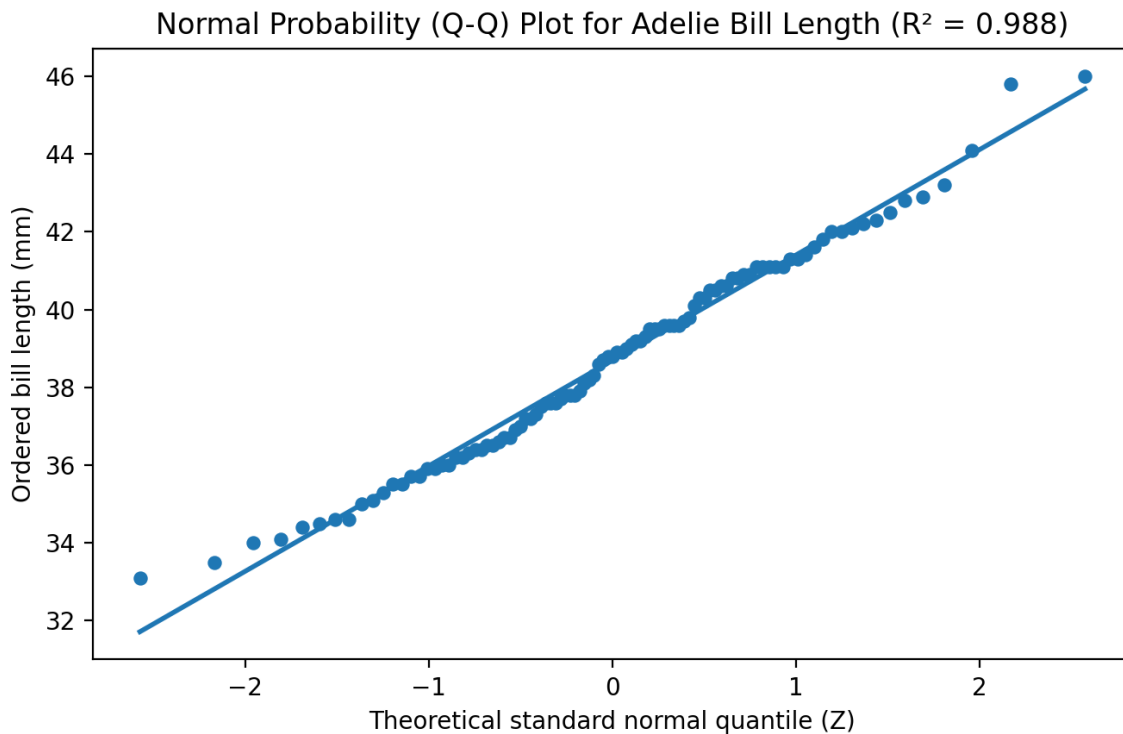
Bill Length has mean 38.6909 mm, median 38.8 mm, skewness 0.1962 and kurtosis -0.3321. The histogram is broadly mound-shaped with only mild asymmetry. The sample IQR is $40.8 - 36.5 = 4.3$ mm, while $1.33s = 1.33 \times 2.7252 = 3.6245$ mm; the IQR is somewhat larger than the lecture benchmark. The range is $46.0 - 33.1 = 12.9$ mm, while $6s = 16.3512$ mm; the observed range is smaller than the rough 6s benchmark. These checks suggest “approximately normal, but not perfect,” rather than exact normality.

Empirical-rule check

Interval	Bounds (mm)	Observed count	Observed %	Normal benchmark
Mean $\pm 1s$	35.9657 to 41.4161	68 / 99	68.69%	68.26%
Mean $\pm 1.28s$	35.2026 to 42.1792	80 / 99	80.81%	80.00%
Mean $\pm 2s$	33.2405 to 44.1413	96 / 99	96.97%	95.44%

These percentages are remarkably close to the normal benchmarks. This provides useful evidence in favour of an approximate normal model, even though the IQR and range checks are not exact.

Normal probability (Q-Q) plot



The Q-Q plot is close to a straight line. The fitted linear trendline has $R^2 \approx 0.988$, which is high. There are modest departures in the extreme tails, but the overall linearity supports using a normal model as a reasonable approximation for introductory probability calculations on this subset.

Fitted normal probabilities versus empirical proportions

Event	Normal-model probability	Empirical proportion	Comparison
$P(X < 36)$	0.1617	$17/99 = 0.1717$	Close
$P(37 \leq X \leq 41)$	0.5341	$47/99 = 0.4747$	Moderate difference
$P(X > 44)$	0.0257	$3/99 = 0.0303$	Close

The fitted normal model reproduces the lower and upper tail proportions fairly well in this subset. The middle interval differs more noticeably. This is a useful reminder that “approximately normal” does not mean that every interval probability will match the sample perfectly.

Z-score check for unusual Bill Length observations

Using $z = (x - 38.6909) / 2.7252$, three observations have $|z| > 2$: 46.0 mm ($z \approx 2.682$), 45.8 mm ($z \approx 2.609$), and 33.1 mm ($z \approx -2.052$). These are unusual relative to the fitted normal centre and spread, but they are not automatically errors. They should be checked against the original record and biological context before any decision to exclude them.

Final interpretation

Taken together, the histogram, small skewness, empirical-rule percentages and Q-Q plot support an approximately normal model for bill length in this 100-record Adelie teaching subset. The model is useful for probability calculations, but the departures in the tails and interval-specific discrepancies show why real data should never be assumed to be perfectly normal.

PART 3 - Pearson MyLab

1. Complete **Quiz #3** on MyLab which has been released on **Tuesday (15-Sep-2026)** and will be **due** by **next Tuesday (22-Sep-2026)** at **5:00pm AWST**.

Note: If you are having issues with obtaining access to MyLab, please re-read the instructions (available in the welcome topic). If you are still unsuccessful, please send an email to the teaching team explaining the issue. It is important that you complete MyLab tasks on a weekly basis since the final e-test in this unit will be completed using MyLab.